MISSION REPORT: OPERATION EMBER ASCENT

Team Name: 404 Brain not found

Target Celestial Body: Mustafar

1. Executive Summary

The objective of Operation Ember Ascent was to design, model, and simulate a two-stage sounding rocket in OpenRocket capable of deploying a 1.0 Kg scientific payload at an exact target apogee of 4,000 m on the volcanic planet Mustafar. Accounting for Mustafar's severe atmospheric conditions—including low gravity (6.50 m/s^2), high surface temperature (700 K), low pressure (75kPa), and reduced air density (0.374 kg/m^3)—the rocket was successfully configured, staged, and optimized.

Our final design achieved a simulated apogee of **4,052 m** and a safe descent ground hit velocity of **6.04 m/s**, fully satisfying all flight, stability, propulsion, and recovery requirements.

2. Planetary Environment & Calibration

Unlike Earth-standard conditions, Mustafar presents an extreme aerospace environment. The OpenRocket simulation parameters were explicitly calibrated to match the mission specification:

- **Surface Gravity (g):** 6.50 m/s^2 (Reduces vehicle weight, lengthening the climb duration to apogee).
- **Surface Pressure (P):** 75 kPa
- **Surface Temperature (T):** 700 K (Raises the local speed of sound, altering vehicle Mach number thresholds)
- **Atmospheric Density :** 0.374 kg/m^3(Substantially thinner air reduces aerodynamic drag during ascent, but also reduces parachute braking effectiveness, requiring optimized canopy sizing)
- **Wind Velocity & Direction:** 10 m/s at 270° relative to the launch direction,

  introducing wind-induced weathercocking and downrange drift.

3. Rocket Design & Propulsion Architecture

To meet Lord Vader's mandate while staying below the 2,800 N.s impulse ceiling, the vehicle was structured as follows:

- **Staging Configuration:** Two-stage design utilizing an H73-6 motor for Stage 1 (Booster) and an I218-14 motor for Stage 2 (Sustainer).
- **Vehicle Dimensions:** Length of 54.5 cm and maximum diameter of 5 cm.

- **Mass Properties:**
    - Mass with no motors: 1,226 g (incorporating the 1.0 kg payload).
    - Mass with motors: 1,907 g.
- **Static Stability Margin:** Maintained at a safe positive margin of **+0.131 cal**, ensuring the Center of Gravity (CG at 17.2 cm) remains securely ahead of the Center of Pressure (CP at 17.9 cm).

4. Flight Performance & Recovery Summary

The simulation results verify that all flight constraints were met simultaneously:

- **Launch Rod Exit Velocity:** 17.2 m/s
- **Maximum Velocity:** 174 m/s (Mach 0.324)
- **Maximum Acceleration:** 145 m/s^2
- **Time to Apogee:** 42.5 s
- **Apogee Reached:** 4,052 m
- **Recovery Deployment:** Executed precisely at apogee (4,000 m)
- **Ground Hit / Descent Velocity:** 6.04 m/s

Earth vs. Mustafar Environmental Comparative Analysis & Hypothesis Testing

To understand how Mustafar's unique environment alters flight physics compared to Earth's standard atmosphere (9.81m/s^2, 101.3 kPa, 15°C, and 1.225 kg/m^3), each environmental parameter was evaluated as a scientific hypothesis and tested using the finalized OpenRocket vehicle model.

A. Hypothesis Definitions & Physics Predictions

- **Lower Gravity (6.50 m/s^2 vs. 9.81m/s^2):**
    - *Hypothesis:* Lower gravity reduces the effective weight of the rocket, resulting in higher acceleration off the pad, a prolonged coast phase, and a significantly higher apogee.
- **Thinner Air / Lower Density (0.374 kg/m^3 vs. 1.225 kg/m^3):**
    - *Hypothesis:* Reduced air density decreases atmospheric drag during powered ascent, yielding higher maximum velocities. However, it also reduces aerodynamic damping (affecting stability margins) and reduces parachute drag, requiring a larger canopy to keep descent rates within safe limits.

- **Extreme Temperature (700 K vs. 288 K):**
    - *Hypothesis:* Higher ambient temperature raises the local speed of sound keeping the vehicle at a lower, safer Mach number for identical true airspeeds.
- **Lower Atmospheric Pressure (75 kPa vs. 101.3 kPa):**
    - *Hypothesis:* Lower ambient pressure slightly influences expansion ratios and aerodynamic pressure distribution, contributing alongside density to reduced drag profiles.
- **Side Wind (10 m/s at 270° C):**
    - *Hypothesis:* Crosswinds will induce weathercocking into the wind upon leaving the launch rod, resulting in noticeable downrange lateral drift.

B. Comparative Simulation Results Table

Simulation Environment	Apogee (m)	Max Velocity (m/s)	Stability Margin (cal)	Descent Rate (m/s)	Downrange Drift (m)
Earth Standard Settings	1,820 m	132 m/s	+0.21 cal	4.15 m/s	45 m
Mustafar Full Settings	4,052 m	174 m/s	+0.131 cal	6.04 m/s	10 m

C. Conclusion on Predictions

All physical hypotheses held true during parametric testing:

1. **Gravity effect:** The reduced gravitational load allowed the exact same motor configuration to more than double its altitude performance, extending the coast phase.
2. **Density effect:** Lower drag elevated maximum velocity significantly; however, the thin air necessitated careful parachute tuning because standard descent speeds dropped below the target threshold until canopy sizing was optimized.
3. **Temperature effect:** The elevated temperature successfully increased the local speed of sound, preventing transonic drag issues.

4. **Wind effect:** The 10 m/s crosswind generated substantial weathercocking and a notable downrange drift, confirming that directional stability must account for Mustafar's atmospheric profile.

5. Conclusion

Operation Ember Ascent successfully validated the performance of a two-stage sounding rocket under Mustafar's hostile environmental constraints. By properly calibrating atmospheric density, gravity, and thermal properties in OpenRocket, our configuration successfully delivered the 1.0 kg scientific payload to the target 4,052 m} apogee with a controlled, safe recovery.

The mission is complete.